

DUMONT D'URVILLE STATION, AntarcticaWMO: **896420**Lat: **66.663S**Lon: **140.001E**Elev: **141**StdP: **14.62**Time Zone: **10.00 (E10)**Period: **94-16**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	-18.9	-16.3	-30.3	1.0	-15.9	-27.7	1.2	-13.1	68.4	8.6	62.9	7.8	14.1	140	1.380

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	6.6	37.5	31.3	35.8	30.4	34.3	29.5	32.3	35.6	31.3	34.5	30.3	33.6	15.3	110

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
30.1	24.3	32.0	28.4	22.5	31.5	26.6	20.7	31.1	12.1	35.9	11.6	34.6	11.2	33.8	37.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 60.8	53.7	47.3	-22.0	40.6	1.7	3.2	-23.2	42.9	-24.2	44.7	-25.1	46.5	-26.4	48.8		
(5)			-21.8	34.1	2.6	1.5	-23.7	35.2	-25.2	36.1	-26.7	36.9	-28.5	38.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	11.9	30.1	23.9	14.6	6.7	2.8	3.3	1.0	1.1	4.3	7.9	19.5	28.6
	DBStd	12.58	3.39	4.49	5.99	6.75	8.81	8.40	8.97	9.38	8.19	6.27	5.64	3.36
	HDD50	13902	616	732	1096	1300	1464	1402	1521	1515	1370	1307	916	663
	HDD65	19377	1081	1152	1561	1750	1929	1852	1986	1980	1820	1772	1366	1128
	CDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD65	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH74	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0
Wind	WSAvg	20.0	17.5	22.4	22.3	21.1	19.9	20.6	19.0	19.5	22.7	19.0	18.4	17.9
Precipitation	PrecAvg	20.0	2.3	2.4	2.1	1.5	1.3	1.3	1.7	1.7	1.5	1.1	1.2	2.0
	PrecMax	27.1	4.7	4.9	3.4	3.1	2.2	3.0	3.2	2.7	2.6	2.0	1.9	4.3
	PrecMin	15.8	0.9	0.8	0.9	0.8	0.5	0.5	0.3	0.9	0.5	0.3	0.6	0.5
	PrecStd	2.8	1.1	0.9	0.7	0.5	0.4	0.6	0.7	0.5	0.6	0.5	0.4	0.8
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	42.0	35.9	30.7	27.0	26.7	23.6	24.2	27.7	27.2	25.5	35.6	39.1
		MCWB	33.4	30.0	28.6	25.6	26.2	21.5	21.7	26.2	25.7	22.3	29.9	32.0
	2%	DB	38.4	33.7	28.3	21.4	22.0	20.3	20.3	22.8	21.8	21.6	32.7	36.7
		MCWB	32.0	28.9	25.8	19.1	19.3	18.1	17.7	21.3	19.1	18.6	27.3	30.7
	5%	DB	36.6	32.4	26.0	18.0	17.8	17.7	16.4	17.9	19.0	19.2	30.3	35.1
		MCWB	31.0	28.1	23.4	15.6	15.6	15.0	14.2	16.5	16.6	16.4	25.3	29.7
	10%	DB	35.2	30.9	23.4	15.6	14.4	15.3	13.5	14.3	16.0	17.0	28.2	33.9
		MCWB	30.2	26.9	20.1	13.3	12.4	13.0	11.6	12.9	14.0	14.4	23.7	29.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	34.7	31.4	29.6	25.8	26.7	21.6	23.2	26.4	25.6	23.1	31.2	33.0
		MCDB	39.2	33.4	30.4	26.8	26.7	23.0	23.8	26.9	26.8	25.9	34.4	37.1
	2%	WB	33.1	30.0	26.3	19.0	19.1	18.6	17.7	21.5	19.5	19.1	28.0	31.6
		MCDB	36.8	32.8	28.0	21.1	21.4	20.0	19.9	22.3	21.2	21.3	32.0	35.0
	5%	WB	32.0	28.8	23.6	15.9	15.8	15.4	14.3	16.5	16.8	16.7	25.7	30.6
		MCDB	35.2	31.8	25.8	17.8	17.6	17.2	15.9	17.8	18.7	18.9	29.7	34.1
	10%	WB	31.0	27.3	20.4	13.6	12.6	12.9	11.8	12.8	14.2	14.4	23.9	29.7
		MCDB	34.3	30.8	23.0	15.3	14.1	15.2	13.6	14.1	16.1	16.7	28.0	33.3
Mean Daily Temperature Range	5% DB	MDBR	6.6	7.6	7.7	7.3	7.6	7.7	8.6	8.6	8.4	9.4	9.8	7.6
		MCDBR	8.0	7.9	7.3	8.4	7.7	8.2	9.9	9.8	9.2	10.1	10.1	8.3
		MCWBR	6.0	6.5	6.5	7.6	7.0	7.5	9.2	8.9	8.3	9.1	8.5	6.4
	5% WB	MCDBR	6.4	6.6	6.6	8.4	7.5	8.0	9.3	9.2	8.8	9.5	9.4	7.5
		MCWBR	5.2	5.9	6.5	7.9	7.1	7.5	8.9	9.1	8.2	8.8	8.2	6.0
Clear-Sky Solar Irradiance	taub		0.261	0.246	0.222	0.178	0.120	N/A	0.121	0.182	0.210	0.221	0.234	0.260
	taud		2.325	2.353	2.328	2.186	1.917	N/A	1.907	2.152	2.251	2.272	2.284	2.283
	Ebn at Noon		317	305	282	229	114	0	102	220	281	313	326	322
	Edh at Noon		35	29	23	13	4	0	3	14	23	31	36	37
All-Sky Solar Radiation	RadAvg		2397	1685	940	343	54	0	19	186	707	1572	2458	2760
	RadStd		124	103	50	13	3	0	1	11	30	60	71	123